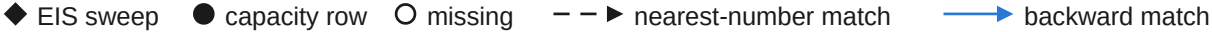
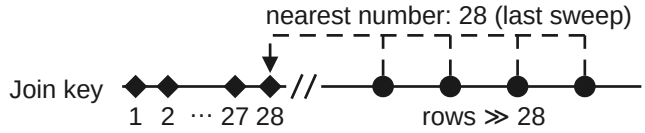
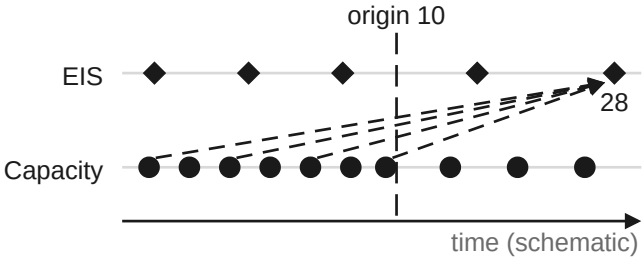


Original join (processed derivative)



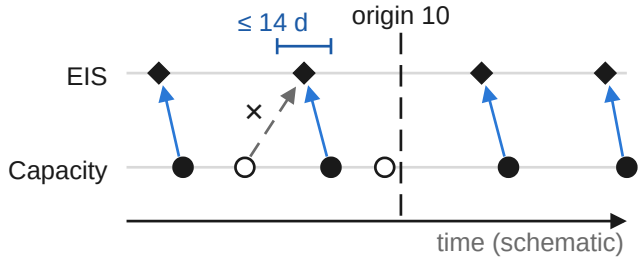


Origin	Rows with future EIS	Max. lead
10	9 / 10	≈ 295 d
20	19 / 20	≈ 85 d

Audited cyclic cell. Audited calendar and profile cells: no future contributors.

Corrected join (this study)

Complete spectra only	Backward in time only
Within 14 days	No match → missing



Outcome

One matched value per cell, all before origin 10, so the benchmark uses capacity only.

(a) Faulty join by row number

(b) Corrected backward join